Process Characterization Plan

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCP-009

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate and complete; AMV, analytical method validation; BLA, Biologics License Application; CCD, central composite design; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; ICH, International Council for Harmonisation; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; PCP, Process Characterization Plan; PCR, Process Characterization Report; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; QbD, quality by design; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SDM, scale-down model; SOP, standard operating procedure; TCID50, fifty-percent tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Small-virus retentive filtration is the dedicated virus removal step of the A-Mab drug substance process, and this plan defines the characterization study that will bound its operating ranges. The step retains virus particles by size, and it is the principal clearance step for minute virus of mice (MVM), the small non-enveloped parvovirus used as the worst-case model virus for the platform. Prior platform experience shows that MVM retention falls as the volumetric load on the filter rises, so the load limit is the quantity that governs the clearance claim. The study will measure that relationship, establish the load limit and the pressure range that supports it, and supply the ranges and the parameter classification that PCR-009 carries into the control strategy. All work will be performed on a qualified small-scale spiking model at the sending site under SOP-1001 and SOP-2012.

The scope is the filtration step as operated on the anion-exchange flow-through pool from Step 8, from the pre-use flush through the post-use integrity test, and it covers the buffer chase that follows the product load, the two model viruses used for the platform claim (MVM and xenotropic murine leukaemia virus, XMuLV), and step yield. Step yield is a process performance attribute and not a quality attribute. The clearance credited to the low-pH hold (Step 6) and to anion exchange (Step 8) is outside this plan. It is characterized in PCR-006 and PCR-008 and consolidated in PCMR-001. Adventitious agent testing, filter supplier qualification and the Stage 2 process performance qualification are also outside the scope of this plan (U.S. Food and Drug Administration 2011).

2 Objectives

The study has five objectives.

1. Measure the effect of volumetric load and filtration pressure on the log reduction achieved for MVM and for XMuLV over the characterization ranges given in §6.
2. Establish the maximum volumetric load at which the step still meets its required log reduction, and confirm that the limit holds at the least favourable pressure.
3. Determine whether step yield and filtrate flux change over the ranges studied, and quantify any change that is found.
4. Determine a proven acceptable range for each parameter against the step acceptance criteria, both with the other parameter at its set-point and with it varying inside its normal operating range.
5. Classify each parameter and define the operating ranges and in-process controls that PCR-009 carries into the control strategy for the transfer to the receiving site.

3 Related documents

This plan sits under the transfer plan PTP-001, the master plan PCMP-001 and the risk assessment RA-001, and its paired report is PCR-009 (Table 3.1).

Table 3.1: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-009	Process Characterization Report (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and the validated analytical methods that govern the study are listed in Table 3.2.

Table 3.2: Controlled procedures and validated methods applicable to this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Small-virus retentive filtration is the last purification operation before formulation, and its only purpose is virus removal. The antibody passes through a membrane whose pore structure retains virus particles by size, so clearance is mechanistic and does not depend on solution chemistry in the way that the low-pH and anion-exchange steps do. Retention is expected to be robust to the composition of the feed delivered by anion exchange. The step is operated in a single pass at constant pressure, and the product load is followed by a buffer chase that recovers antibody held up in the device. It provides the largest single MVM reduction in the train and a substantial XMuLV reduction. No clearance of host cell protein, DNA or leached Protein A is claimed for this step.

The filter type proposed for A-Mab is a platform device that has been used at commercial scale for three prior humanized IgG1 products (X-Mab, Y-Mab and Z-Mab). Characterization of those products established the two behaviours that shape the present study. First, the log reduction achieved for MVM falls as the volumetric load rises, because retention capacity is consumed as the membrane is loaded. Second, XMuLV is retained with a wide margin at every load examined, since the enveloped particle is much larger than the retention rating of the membrane. This prior knowledge can be applied to A-Mab because the removal mechanism is size exclusion, the device is unchanged, and the feed presented to the filter is a polished flow-through pool of comparable composition and turbidity. It is not sufficient on its own. The load limit is product specific, A-Mab has not previously been filtered at the loads proposed for commercial operation, and the limit is therefore established here.

Pressure sets the flux and the time the product spends on the membrane. Platform data show only modest flux decay across the load range of interest, which suggests that pore plugging is limited and that pressure has little influence on retention. Pressure is nevertheless carried into the design. An excursion above the operating limits of the device could deform retained particles or the membrane itself, and the consequence would be a loss of retention at the condition where the margin is already smallest, so the study is designed to bound that possibility and not to assume it away.

4.2 Quality attributes in scope

The step governs the two viral clearance attributes of the drug substance, and both are of very high criticality (Table 4.1). Their acceptance criteria apply to the purification train as a whole and not to any single step, so this step's own requirement is obtained by subtracting the clearance credited to the other steps of the modular claim (International Council for Harmonisation 2023). The step requirement for each virus is stated in §7.2. MVM carries the lower cumulative requirement of the two, and it is the harder of the two to meet, because the parvovirus is close to the retention rating of the membrane and the low-pH hold contributes nothing to its clearance.

Table 4.1: Quality attributes governed by the small-virus retentive filtration step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

No product-quality attribute is formed or cleared at this step, so none is placed in scope. The 3 responses measured in the study are the two log reduction factors and the step yield.

4.3 Risk-based prioritization of parameters

RA-001 assessed the parameters of this step before any characterization work and assigned each one a study type. Two parameters were ranked high enough to require multivariate evaluation, the volumetric load and the filtration pressure, and they are the 2 factors of the design described in §6. The load was ranked on the failure mode that matters most at this step, which is a volumetric load above the routine limit and the reduced MVM log reduction that follows from it. Pressure was ranked lower on severity but retained in the design, because an excursion outside the operating limits of the device could affect virus removal at high load, and because its interaction with load cannot be assessed by varying one parameter at a time.

The existing controls recognized in RA-001 are the volumetric load limit applied to the product load and the chase together, the pressure limits of the device, and the pre-use and post-use integrity tests (SOP-2012). These controls are the reason the step has operated without incident on the platform. They are not evidence for the ranges proposed for A-Mab, and the study is what supplies that evidence.

The remaining operating conditions of the step are fixed by SOP-2012 and are treated as controlled inputs. They include the filter type and effective area, the pre-use flush, the buffer used for the chase, and the operating temperature of the suite. Each is verified before every run and recorded, so that a change in any of them is visible in the study record.

5 Materials and methods

5.1 Scale-down model and its qualification

The study depends on a small-scale model that behaves like the commercial device, and qualifying that model is the first activity of the plan. The model will use the same membrane chemistry, the same filter construction and the same lot family as the commercial device, at a reduced effective filtration area. Scaling is by area. The volumetric load, expressed per unit area, and the filtration pressure will be held identical to the values used at scale, so that flux and load per unit membrane are preserved. Qualification will follow SOP-1001 and will compare the scale-down device with commercial-scale batches at the centre condition on filtrate flux decay, step yield and post-use integrity. The acceptance criteria for that comparison are given in §7.1.

Viral clearance studies are performed on spiked material, and the spike itself can compromise the study if it is not controlled (International Council for Harmonisation 2023). The virus stock will be filtered before it is added, so that aggregates and cell culture debris carried in the stock do not foul the membrane and produce a clearance result that the unspiked process would not reproduce, and the filtered stock will be titrated before use. The spike will be added as a small volume fraction of the load, and the spiked load will be assayed after mixing to establish the actual challenge titre. An unspiked control run at the centre condition will be executed alongside the spiked runs, and it provides the flux and yield reference for the spiked material. The replicate centre runs of each design provide the estimate of pure error that the lack-of-fit test needs.

5.2 Operation

The step will be operated as it is at commercial scale, under SOP-2012. The feed is the anion-exchange flow-through pool of Step 8, adjusted to the platform protein concentration and passed through the prefilter specified in the operating procedure (SOP-2012). Filtration proceeds in a single pass at constant pressure until the target volumetric load has been delivered, and the product load is then followed by the buffer chase. The volumetric load recorded for each run is the product load and the chase together, which is the quantity the control strategy limits. A post-use integrity test will be performed on every device. A run whose device fails the post-use test is invalid, and it will be repeated on a new device once the cause has been established.

5.3 Analytical methods

Two validated infectivity assays supply the clearance responses of the study (Table 5.1). MVM is titrated by TCID50 with quantitative PCR confirmation under AMV-3018, and XMuLV is titrated by TCID50 under AMV-3017. The log reduction factor for each run is calculated from the total virus in the spiked load and the total virus in the pooled filtrate (the chase included). Assay precision, the limit of detection and the qualification of the cell lines used for titration are reported in the method validation packages listed in Table 5.1, and they are not repeated here.

Table 5.1: Validated analytical methods for the study responses.

Reference	Title	Type
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation

Where no virus is recovered from the filtrate, the log reduction factor is reported as a lower bound set by the detection limit of the assay, and the study will claim no more than that bound. Step yield will be calculated by mass balance from the protein concentration of the load and of the pooled filtrate, measured by the platform ultraviolet absorbance method referenced in PCMP-001.

5.4 Sampling plan

Sampling is designed so that clearance can be reported as a function of load and not only at the end of a run. The spiked load pool will be sampled after mixing and before filtration. The filtrate will be collected in sequential fractions across the volumetric load, and each fraction will be assayed for both model viruses (MVM and XMuLV). A pooled filtrate sample, made by combining the fractions in proportion to their volume, gives the log reduction factor reported for the run. The fraction data give the cumulative log reduction at every load between the start of the run and the maximum load studied, so the load at which clearance begins to fall can be read directly from the fractions and does not depend only on the fitted model. Samples from the unspiked control run will be retained for yield and for the filtrate quality checks named in SOP-2012.

The screening design includes 3 centre-point runs and the response-surface design includes 4, and those replicates carry the pure error estimate for both designs.

5.5 Statistical methods

Analysis will follow SOP-1002. Each factor will be coded so that the coded levels -1 and $+1$ are the edges of its characterization range and 0 is the midpoint. The design centre is the midpoint of the characterization range, at 95 L/m2 and 19 psi, and it is not the set-point of either parameter. Performance at the set-point will be predicted from the fitted model and not read off the centre runs.

The screening data will be fitted by ordinary least squares as a model in both main effects and their interaction, with the effect of a term reported as twice its coded coefficient, and the response-surface data will be fitted as a full quadratic model in the same 2 factors. Terms will be judged at $\alpha = 0.05$. Model adequacy will be judged on R^2, adjusted R^2, predicted R^2 from the PRESS statistic, and the analysis of variance partition of the residual into lack of fit and pure error. A response for which no term is significant will be reported as robust over the ranges studied, and no predictive model will be claimed for it. The screening design identifies which effects are active. The response-surface model is the predictive model, and the operating region, the proven acceptable ranges and the load limit will all be derived from it.

Commercial-scale capability will be estimated by Monte-Carlo simulation of 2,000 batches with both parameters sampled inside their normal operating ranges (Table 6.1), and reported as a one-sided Cpk against the step acceptance criterion of §7.2.

6 Study design

6.1 Factors, ranges and study type

The design covers the two parameters that RA-001 assigned to multivariate study, over ranges wider than routine operation (Table 6.1). The coded letters used in the design matrices and in the analysis are given in Table 6.2.

Table 6.1: Parameters to be studied, with set-points, characterization ranges and normal operating ranges.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Filtration volume (load)	L/m2	90	50–140	50–105	multivariate
Filtration pressure	psi	13	8–30	10–20	multivariate

Table 6.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

Volumetric load will be studied from 50 to 140 L/m2, and the upper edge is 1.33 times the upper edge of its normal operating range. The range extends upward only. Risk to retention increases with load, and a load below the routine range offers no mechanism by which retention could be lost, so widening the range downward would consume runs without adding knowledge. The upper edge was chosen so that the study can locate the load at which the acceptance criterion is no longer met, if that load lies within reach of the operation proposed for commercial manufacture.

Filtration pressure will be studied from 8 to 30 psi, a span 2.2 times the width of its normal operating range, and it extends on both sides of that range. Low pressure lengthens the filtration and is the condition under which a slow loss of retention would appear. High pressure is the condition under which the membrane and the retained particles are most stressed. Both edges were set within the pressure limits recorded for the device in SOP-2012. The set-point of each parameter lies inside its normal operating range and inside the characterization range (Table 6.1).

6.2 Screening design

The screening design is a two-level full factorial in the two factors with 3 centre-point replicates, 7 runs in total (Appendix A). With only two factors the design is not fractional, so both main effects and the interaction are estimated free of aliasing. The centre replicates supply the pure error estimate and a test for curvature. Runs will be executed in randomized order, and the run order will be recorded with the data. The purpose of the screening stage is to identify which effects are active and to confirm the direction and the approximate size of the load effect before the response-surface runs are committed.

6.3 Response-surface design

The response-surface design is a face-centred central composite design in the same two factors, 12 runs in all. It comprises 4 factorial runs, 4 axial runs at the faces of the characterization range and 4 centre replicates (Appendix B). The axial runs add the quadratic terms. Curvature is what the load limit depends on, because a limit read from a straight line fitted between two extremes would misplace the load at which clearance falls below the criterion. The worst case for virus retention is the highest load, and the design executes that corner directly at both edges of the pressure range. No separate confirmation run at the worst-case corner is needed, because the corner is a design point in both stages of the study.

6.4 Univariate assessment

No parameter of this step is assigned to univariate study. RA-001 identified two parameters for multivariate evaluation, and the remaining operating conditions are held at their platform values and verified as inputs (§4.3). Two supporting evaluations will nevertheless run alongside the designed experiments. The first is the unspiked control at the centre condition, which provides the flux and yield reference described in §5.1. The second is the post-use integrity test on every device, which is the direct check that the membrane was intact for the run whose clearance is being claimed.

7 Acceptance and decision criteria

The criteria below are fixed before the data exist, so that the outcome of the study is decided by rule and not by inspection (U.S. Food and Drug Administration 2011).

7.1 Scale-down model qualification

The scale-down model is qualified when its filtrate flux decay profile and its step yield at the centre condition are not significantly different from the commercial-scale reference under the test described in §5.5, and when every device used in the comparison passes its post-use integrity test. A model that fails either criterion will be modified and requalified before any spiked run is executed, and the modification will be described in PCR-009.

7.2 Clearance acceptance

The acceptance criterion for this step is a step-level log reduction, back-calculated from the cumulative requirement for the drug substance minus the clearance credited to the other steps of the modular claim (International Council for Harmonisation 2023). For MVM the cumulative requirement is 8.6 $\log_{10}$, of which 4.71 $\log_{10}$ is credited to the other steps, so this step must achieve at least 3.89 $\log_{10}$. For XMuLV the same calculation gives a step requirement of 3.65 $\log_{10}$ against a cumulative requirement of 16.7 $\log_{10}$. Every run must meet the step criterion for both viruses, the criterion must also be met by the model prediction across the whole of the region carried forward, and a run that fails it will be investigated before the design is analysed. The clearance claimed in PCR-009 will be the lower of the value demonstrated at the worst-case corner and the value predicted at the maximum load, and it will not exceed the bound imposed by the detection limit of the assay.

7.3 Model adequacy

A response-surface model will be used for prediction only when its lack-of-fit test is not significant at the level stated in §5.5 and its adjusted and predicted R^2 support prediction inside the design region. Where a response shows no significant term, it will be reported as robust across the ranges studied, with those ranges stated, and no model will be fitted for prediction. A robust response is a result of the study and is carried into the knowledge space on that basis.

7.4 Load limit and operating region

The operating region is the set of load and pressure combinations at which both step criteria are predicted to be met while the other parameter varies inside its normal operating range. The maximum volumetric load carried into the control strategy will be the largest load at which the MVM criterion is met under that analysis, capped at the upper edge of the characterized range (Table 6.1). The worst case the region must satisfy is the maximum load combined with the least favourable pressure, and the region will be declared only if that corner meets both criteria. If the MVM criterion fails at a load inside the normal operating range, the normal operating range will be reduced to the demonstrated limit. That outcome would be recorded as a deviation from the assumptions of this plan, and PCR-009 would state the reduced limit and its consequence for the modular clearance claim.

7.5 Capability

Capability will be estimated as described in §5.5 and judged against the minimum Cpk set out in the acceptance framework of PCMP-001. No numeric capability threshold is fixed in this plan, because the threshold is a campaign-level criterion and is stated once for the whole campaign in the master plan.

8 Proven acceptable ranges (planned analysis)

The operating region of this step is multivariate, and the proven acceptable ranges complement it with a statement per parameter that an assessor and an operator can both use. A range will be determined for each parameter against each of the two clearance criteria, and reported together with the analysis that produced it.

The acceptance basis is the step-level criterion of §7.2 for both viral responses. Step yield maps to no drug-substance criterion, so no proven acceptable range is determined for it.

Two analyses will be run for each parameter and each viral response. The first holds the other parameter at its set-point and scans the parameter of interest across its characterization range on a grid of 81 points, using the fitted response-surface model. The second propagates the normal operating range of the other parameter. At each grid point the other parameter is drawn 2,000 times from within its normal operating range, and the response is predicted for every draw. The range reported by the second analysis is one that survives routine variation in the parameter that is not being scanned, which is the form a control strategy can rely on.

The proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the response stays within acceptance under the analysis in question. Where the two analyses give the same interval, and that interval is the whole characterization range, the parameter is proven acceptable across everything studied and is robust to variation in the other parameter. Where the propagated analysis gives the narrower interval, PCR-009 will name the binding response and give the margin that remains at the set-point.

Each range will be reported with a plot of the response against the parameter, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked, and the two analyses appear as the two panels of that plot. The ranges are statements about one parameter and the operating region is multivariate, so where a whole characterization range is proven acceptable the binding constraint on operation is the corner of the region and not the range. Every range will be bounded by the characterization ranges of §6.1 and by the fitted model, and none will be extrapolated beyond a studied edge.

9 Data management and integrity

Study data will be recorded and retained under the ALCOA+ principles. Raw analytical data will be captured in the validated laboratory information management system (LIMS), run records in the electronic notebook, and the statistical analysis in a version-controlled script that regenerates every table of PCR-009 from the raw data. All entries are attributable, contemporaneous and reviewed by a second person before the data are used in the analysis. A change made after review requires an audit trail entry with a reason.

10 Roles and responsibilities

The functions responsible for the study and for its report are listed in Table 10.1. Process Development owns the plan, its execution and the report. Quality Assurance approves the plan before execution and approves PCR-009 on completion.

Table 10.1: Functions and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Authors the plan; qualifies the scale-down model; executes the designed experiments; authors PCR-009
Analytical Development	Performs the infectivity assays and the protein assay; maintains the method validations
Virology laboratory	Prepares and titrates the virus stocks; performs the spiked runs under containment
Statistics	Reviews the designs before execution; performs and reviews the model fitting and the capability estimate
Manufacturing Science	Reviews the scale-down qualification and the ranges proposed for the receiving site
Quality Assurance	Approves this plan and PCR-009; dispositions any deviation raised during the study

11 Deliverables and schedule

The deliverable of this plan is PCR-009, the characterization report for this step. The study runs in four phases, which are the scale-down model qualification, the screening design, the response-surface design, and the analysis and reporting. Each phase begins only when the preceding phase has been reviewed, and the response-surface runs are committed only after the screening results have been assessed against §7. No calendar dates are fixed in this plan. The campaign schedule is held in PCMP-001.

12 Risks and assumptions

The principal risk to the study is that the scale-down model does not represent the commercial device. The model is qualified before use (§5.1), but the qualification is performed at the centre condition, and representativeness at the extremes of the load range is an assumption the study cannot itself test, so PCR-009 will state that bound on its conclusions.

The second risk is the virus spike. A spike carrying aggregated material can foul the membrane and depress the apparent clearance at high load, which would place the load limit lower than the unspiked process warrants. Filtration of the virus stock before addition and the unspiked control run are the controls for that risk (§5.1).

The third risk is analytical. The largest log reduction the study can demonstrate is fixed by the titre of the spiked load and by the detection limit of the assay, so a result at that bound is a property of the method and not a measurement of the filter. Clearance will therefore be claimed only to the bound, as stated in §5.3.

The study assumes that the anion-exchange pool used as feed is representative of routine operation, and that the filter devices come from a single lot family. Both assumptions will be recorded in PCR-009 with the identifiers of the material actually used. A change of membrane lot family during the study would require the affected runs to be repeated.

13 Approvals

This plan becomes effective when it has been approved by the functions listed in Table 1. Execution will not begin before that approval has been recorded.

Appendix A — Planned screening design matrix

The coded screening design is given in Table 13.1. Response columns are not shown, because no data exist when this plan is approved.

Table 13.1: Planned screening design matrix, coded levels (A = filtration volume, B = filtration pressure).

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	center	0	0
6	center	0	0
7	center	0	0

Appendix B — Planned response-surface design matrix

The coded response-surface design is given in Table 13.2 and the same design in natural units in Table 13.3.

Table 13.2: Planned response-surface design matrix, coded levels.

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	axial	-1	0
6	axial	1	0
7	axial	0	-1
8	axial	0	1
9	center	0	0
10	center	0	0
11	center	0	0
12	center	0	0

Table 13.3: Planned response-surface design matrix, natural units.

Run	Type	Filtration volume (load)	Filtration pressure
1	factorial	50	8
2	factorial	50	30
3	factorial	140	8
4	factorial	140	30
5	axial	50	19
6	axial	140	19
7	axial	95	8
8	axial	95	30
9	center	95	19
10	center	95	19
11	center	95	19
12	center	95	19

References

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U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.